Usage of AI Tools

While populating data in Oracle Database, we use ChatGPT to assign different values in each tables as the graph provided below.

```
--Shift Table

CREATE TABLE SHIFT (
Shift_ID INTEGER PRIMARY KEY,
Shift_Date DATE,
Start_Time TIMESTAMP,
End_Time TIMESTAMP,
Assigned_Volunteers VARCHAR2(100)
);

--Insert statement
INSERT INTO SHIFT (Shift_ID, Shift_Date, Start_Time, End_Time,
Assigned_Volunteers)
VALUES ();

populate 10 rows of data
```

Here are 10 rows of sample data for the `SHIFT` table:

```sql
-- Insert statements for SHIFT table
INSERT INTO SHIFT (Shift_ID, Shift_Date, Start_Time, End_Time, Assigned_Volunteers)
(1, DATE '2024-08-01', TIMESTAMP '2024-08-01 08:00:00', TIMESTAMP '2024-08-01 12:00
(2, DATE '2024-08-02', TIMESTAMP '2024-08-02 12:00:00', TIMESTAMP '2024-08-02 16:00
(3, DATE '2024-08-03', TIMESTAMP '2024-08-03 08:00:00', TIMESTAMP '2024-08-03 12:00
(4, DATE '2024-08-04', TIMESTAMP '2024-08-04 12:00:00', TIMESTAMP '2024-08-04 16:00
(5, DATE '2024-08-05', TIMESTAMP '2024-08-05 08:00:00', TIMESTAMP '2024-08-05 12:00
(6, DATE '2024-08-06', TIMESTAMP '2024-08-06 12:00:00', TIMESTAMP '2024-08-06 16:00
(7, DATE '2024-08-07', TIMESTAMP '2024-08-07 08:00:00', TIMESTAMP '2024-08-07 12:00
(8, DATE '2024-08-08', TIMESTAMP '2024-08-08 12:00:00', TIMESTAMP '2024-08-08 16:00
(9, DATE '2024-08-09', TIMESTAMP '2024-08-09 08:00:00', TIMESTAMP '2024-08-09 12:00
(10, DATE '2024-08-10', TIMESTAMP '2024-08-10 12:00:00', TIMESTAMP '2024-08-10 16:0
```

Table of Contents

Executive Summary

While the Olympic Games are undoubtedly a “major sportive and cultural feast” (Pop, 2013), the general public often overlooks the enormous effort involved behind the scenes. This year’s Paris Olympics has over 45,000 volunteers contributing to the organisational challenge (Olympics, 2024), meaning it is integral for the Organising Committee to proactively take steps to maximise efficiency. This report presents a proof-of-concept project for a Volunteer Management System in the 2024 Paris Olympics. As a business database consulting group, we seek to create a database system to facilitate efficient coordination and communication for a successful event.

The Entity Relationship Model (ERM) will map out the real-world implications of the system. A comprehensive examination of the stakeholders involved and the links between them will be conducted using seventeen entities. This report will justify the use of these entities and any assumptions involved. From this model, we will utilise data warehousing techniques to create a star schema, while illustrating how fact and dimension tables were built. Finally, using the materialised view in conjunction with visualisations, we will utilise PowerBI through event based analysis and task based analysis to gain valuable, data driven insights into the running of the Games.

Through ERM creation, data warehousing and data analysis, we seek to initiate a solid foundation for the Volunteer Management Program, ultimately contributing to the success of the Olympic Games.

ERM Scope and Assumptions

Scope

As this database aims to manage volunteers, there are many features and components of Olympic management that are out of the defined scope. This includes venue security measures, cybersecurity measures, international broadcasting logistics, international travel arrangements, and accommodation logistics. This was a conscious decision, to ensure that the purpose of the project was met.

Stakeholder Analysis

Stakeholder	Requirements
Olympic Organising Committee	The Olympic Organising Committee oversees the execution of the Olympic Games, from inception to completion. Their requirements include volunteer allocation and management, along with resource management and integration with pre-existing systems to ensure usefulness (Ribeiro et al., 2021).
Staff and Event Managers	Staff and Event Managers have similar high-level requirements but are focused on more specific tasks. In the context of volunteer management, they require information related to scheduling, training and incident monitoring to connect information with other services and databases (Dixon, 2023).
Volunteers	Volunteers require access to information, training, and services to complete tasks effectively. This entails the management of personal information, training, shift scheduling, and safety protocol information.

Supertype and Subtype

The supertype is identified as “Services”, with two subtypes, including “Health Services” and “Transport Services”. Common attributes to both services are added to the supertype, namely Service ID, Location ID, Service Description, Operating Hours, Service Type, and Emergency Contact. Unique to the Health Service subtype is resources, and unique to the Transport Service subtype is route number.

Assumptions

- Volunteers must complete mandatory training for their roles
- Volunteers can be associated with multiple events, roles, shifts, and training sessions
- Shifts can span over different time periods

- Shifts are directly related to volunteers, as they indicate the times when a volunteer is needed
- Each role has specific skills and training requirements
- Roles are associated with tasks that can be assigned to volunteers or staff
- Each department oversees specific tasks, protocols and staff
- Protocols define the procedure and compliance requirements linked to each department
- Staff are differentiated from volunteers, as they are in a paid role
- Staff are assigned to departments, linked to completing certain tasks within their roles
- Events are associated with specific locations, with each location carrying a varied capacity amount
- Spectators are linked to events that they attend
- Operational support is linked to events, providing support in an organisational or logistical capacity
- Health services are provided at specific locations, for spectators, athletes, volunteers, and staff if required
- Transport services are linked to certain locations, facilitating the movement of spectators, athletes, volunteers, and staff if required

Justification

Entity	Justification
Team	This entity was included to link to the "athlete" entity and to provide organisers with the ability to see where all volunteers, staff, athletes, and teams would be on a given event day.
Athlete	Over "11,000 athletes" (Suliman & Granados, 2024) are in the 2024 Paris Olympics, highlighting the need to manage them all. One team often includes many athletes. In fact, there are over 460 athletes in Team Australia in the 2024 Paris Olympics.
Event	The Olympics hosts many events, with the 2024 Paris Olympics holding "329 medal events across 45 sports" (Leuzzi, 2024). Volunteers are required to keep these events running smoothly.
Operational Support	This entity is linked to events, providing logistical and technical assistance. Volunteers may have operational support tasks within their roles or tasks if an issue arises.
Spectator	Many spectators attend the Olympics, with the recent Paris Olympics expecting "326,000" spectators (BBC, 2024). It is important to know how many spectators are expected at events to ensure the capacity limit is not reached once volunteers are scheduled.
Volunteer	This entity captures data related to the volunteers, such as names, contact details, and DOBs, ensuring they can be scheduled for tasks and roles and that they receive certificates after volunteering.

Shift	Shifts denote when the volunteer will be scheduled to work and complete their tasks. One shift can have many volunteers, as the Olympics is a large-scale event.
Training Session	Training is crucial for volunteers to participate in their roles effectively. Volunteers have a series of intricate tasks, all of which they need training and guidance to learn how to do. One training session can be completed by many volunteers, as it is a collaborative group experience.
Role	The role entity defines the position of each volunteer at the Olympics, linked with skills and training to ensure they are able to complete tasks effectively. Each role can be assigned to multiple volunteers, based on venue capacity and requirements and can encompass multiple tasks.
Task	The task entity denotes the actions within each role and helps allocate responsibilities. Each role can contain multiple tasks, and each staff member can also be assigned multiple tasks, similar to volunteers.
Staff	This entity contains information regarding staff, including their unique identification, task description and hours. The system utilises it to monitor the progress of individuals and differentiate between them.
Department	The department entity is useful in grouping staff and volunteers. It enables insights into shared tasks and responsibilities, allowing for streamlining of processes.
Protocol	The protocol entity entails the various levels of compliance that each volunteer or staff member must adhere to, informing the allocation of tasks.
Location	Location is important to include as despite the Olympics being held in one city, events are often held in many locations. For example, the 2024 Paris Olympics has events scattered throughout the city (Contreras, 2024).
Services	Services are included as a supertype. These services are essential to volunteers, and one location may have many different types of services.
Health Services	Health services have the unique attribute of available resources, to ensure that organisers can see what is available and potentially send volunteers to deliver equipment if necessary.
Transport	Transport services have the unique attribute of route number to

Services	ensure that volunteers are headed to the correct location, and are given adequate transport to complete their tasks.

Conversion of Entities

Step 1: Get Rid of The Unneeded Attributes

- How we identified unneeded attributes
  - Whether it can ensure the management is smooth of such a big volunteer group.
- What we identified as unneeded attributes
  - Assigned_Volunteers, Training_Location, Role_Description, Required_Training, Location_ID, Location_Name, Accessibility_Features (Further explanation is provided in next section)

Step 2: Create OLYMPIC_VOLUNTEER_FACT Table

- Why we identified volunteer entity as a fact table
  - The Organising Committee of the 2024 Olympic Games decided to develop a volunteer management system.
- What is contained in our OLYMPIC_VOLUNTEER_FACT TABLE
  - Keys + Volunteer_First_Name + Volunteer_Last_Name

Step 3: Start Creating Simple Dimension Tables

- What we defined as simple dimension tables
  - Entities that do not include Foreign Keys
- In our case, they are
  - Shift and Training

Step 4: Analyse FKs to Know What Entities Can Be Combined Into One Dimension Table

- We went through the ERM multiple times to have a deep understanding of the relationships between each entity.
- The combinations we get
  - Role + Task
  - Event + Spectator + Operational Support
  - Location + Health Service + Transport

Step 5: Create Dimension Tables

- ROLE_DIMENSION= Role + Task
- EVENT_DIMENSION= Event + Spectator + Operational Support
- LOCATION_DIMENSION= Location + Health Service + Transport

Star Schema Construction

SHIFT Dimension Table

Included	
Shift_Key	Was Shift_ID in the SHIFT TABLE.
Shift_Date	Included for scheduling purposes.
Start_Time	Defines working hours for each volunteer.
End_Time	
Not Included	
Assigned_Voulunteers	Not included because the volunteers are likely to be assigned to different tasks frequently but the dimension table is typically used for categorical information which is less likely to change frequently.

TRAINING HISTORY Dimension Table

Included	
Training_History_Key	Was Training_History_ID in the TRAINING HISTORY TABLE.
Training_Type	Included to make it easier to categorise different types of history training sessions.
Training_Date	Included for scheduling purposes.
Training_Status	Status of the training included 'Completed' or 'Scheduled', which is useful to track the progress and completion of training sessions.
Duration	Shows the length of the training,
Not Included	
Training_Location	Not included because the volunteers are likely to be allocated to different locations based on different training types, which are likely to be changed frequently.

ROLE Dimension Table

Included	
Role_Key	Was Role_ID in the ROLE TABLE.
Role_Name	Necessary to identify the role and responsibilities of the volunteer.
Skills_Required	Included to help match volunteers with related roles based on their skill sets.
Task_Name	Specifies what the volunteers should do.
Completion_Status	Included for tracking purposes. 'Completed', 'In Progress', or 'Pending' tracks whether the task is completed or not.
Not Included	
Role_Description	Not included because Role_Name provides an understanding of what the role is about to do.
Required_Training	Not included because training requirements are context-specific and can vary between different events or tasks. Therefore, it is difficult to manage in the dimension table.

EVENT Dimension Table

Included	
Event_Key	Was Event_ID in the EVENT TABLE.
Event_Name	Necessary to identify the events that the volunteers are participating in.
Event_Date	Captures the date of the event when volunteers process their tasks.
Event_Type	Defines the type of the event, helping in filtering and analysing.
Event_Capacity	Indicates the maximum number of attendees that the event can accommodate.
Spectator_First_Name	Identifies individual spectators. Splitting the last name and first name into two columns increases readability and makes searching easier.
Spectator_Last_Name	
Support_Type	Defines the type of support provided during the event, helping to manage the resources fluently.
Not Included	
Location_ID	Not included since the EVENT DIMENSION TABLE only focuses on attributes that are inherent to the event.

LOCATION Dimension Table

Included	
Location_Key	Was Location_ID in the LOCATION TABLE.
Location_Address	Included to emphasise the physical address of the location.
Location_Capacity	Indicates the maximum number of attendees that the event can accommodate, would be essential in safety regulations and event planning.
Health_Service_Type	Captures the type of health services available at the location for either attendees, staff, or volunteers. Helps in planning safety measures.
Available_Resources	Captures the available resources at the location, helping in assessing whether the events are suitable to target attendees.
Transport_Type	Describes the types of transportation available for attendees or volunteers to reach the location.
Emergency_Contact	Essential for all people involved to gain quick response and rescue in case of emergencies.
Not Included	
Location_Name	Not included because this is similar to the location address.
Accessebility_Features	Since not all locations have accessibility features, and most locations did get the same accessibility features, this column is not included.

VOLUNTEER Fact Table

Included	
Volunteer_Key	Identifies the volunteer information.
Training_History_Key	Identifies the training history information.
Event_Key	Identifies the event information.
Role_Key	Identifies the role information.
Location_Key	Identifies the location information.
Volunteer_First_Name	These columns identify individual volunteers. Splitting the last name and first name into two columns increases readability and makes searching easier.
Volunteer_Last_Name	

PowerBI Analysis

Volunteering provides numerous benefits both to the atmosphere and diversity of the Olympic Games, but also to volunteers themselves. Hence, it is integral to utilise data analysis when exploring management systems. PowerBI allows us to import data from the materialised view and create visualisations that provide insights into various aspects such as resource management and performance analysis.

Event Based Analysis

Event-based analysis allows us to determine which aspects of the Olympic Games are the most resource-intensive, enabling the volunteer management program to allocate resources effectively.

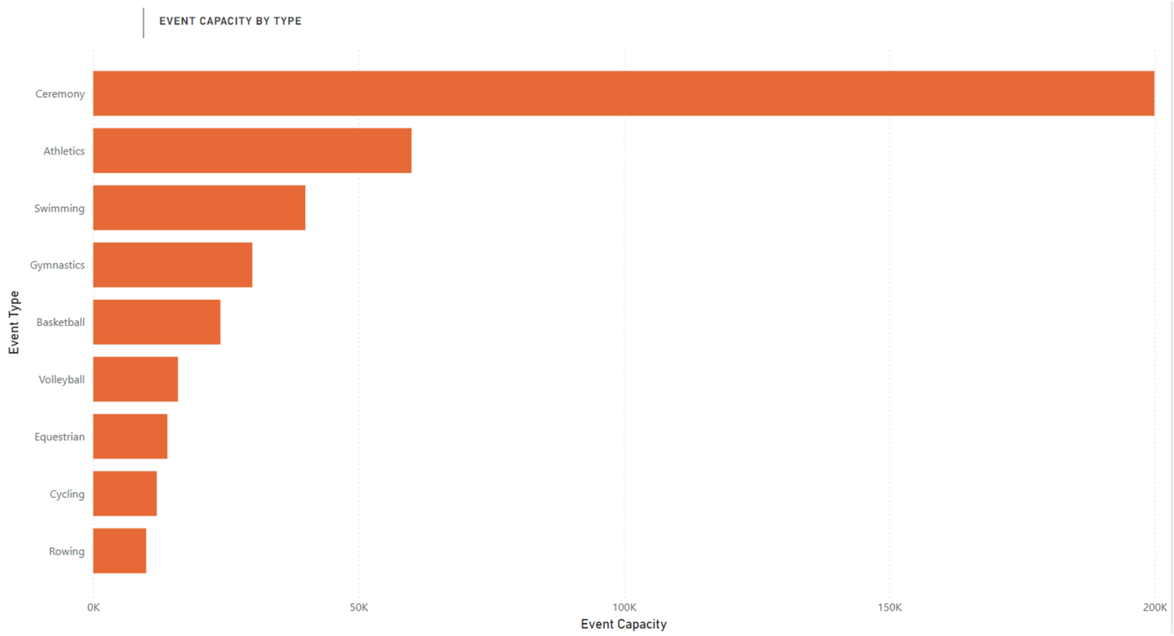


Figure 3.1: Bar Chart of Event Capacity by Type

The Olympic Games are made up of a vast array of events with each having its own requirements. Figure 3.1 provides insights into the events with the largest capacity, allowing us to understand the relative distributions and sizes of different events. It reveals ceremonies by far require the highest capacity, as logically expected since all Olympians would be present. More interestingly, the remainder of the visualisation depicts a list of sports sorted by event capacity, providing insights into which events draw the largest crowds or have the largest stadiums, and assisting in tailored resource allocation. The volunteer management system can optimise the distribution of volunteers and facilities adequately to cater for event capacities and attendance trends.

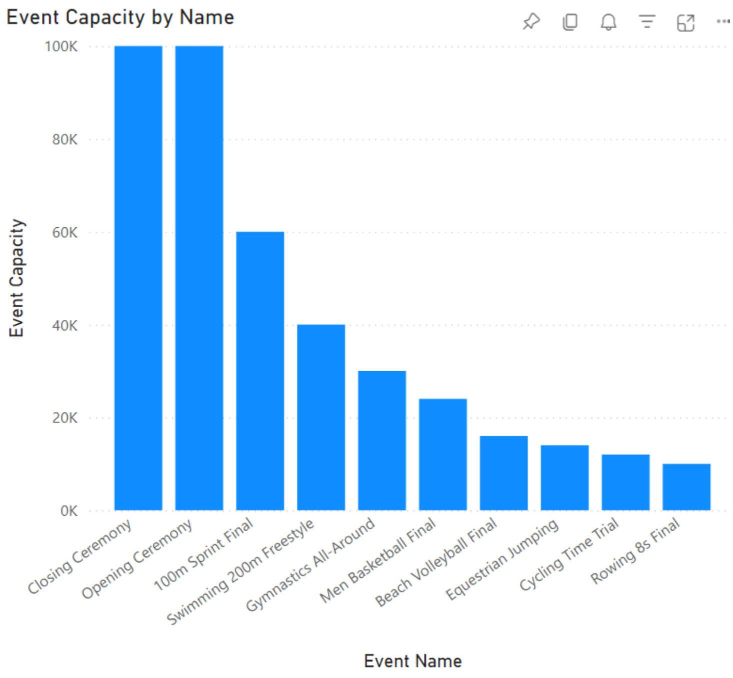


Figure 3.2: Column Chart of Event Capacity by Name

Whilst event-based analysis by sport is helpful, a deeper analysis of event capacities allows us to target resources. Figure 3.2, aligning with our findings from Figure 3.1, shows that the opening and closing ceremonies are by far the most resource-intensive events, followed by the 100m Sprint and 200m Freestyle. These findings can help ensure that the management system accounts for and is prepared for high-capacity events, optimising resources and support services. Visualisations can communicate event performance to stakeholders or other strategic bodies, whilst helping track progress towards goals such as increasing attendance.

Task Based Analysis

COMPLETION_STATUS	TASK_NAME	EVENT_NAME	ROLE_NAME	SUPPORT_TYPE
Pending	Set Up Refreshments	Beach Volleyball Final	Refreshment Setup	Cleaning
Pending	Collect Feedback	Closing Ceremony	Feedback Collector	Information
Pending	Clean Up Venue	Cycling Time Trial	Cleanup Crew	Transportation
Pending	Check-In Attendees	Swimming 200m Freestyle	Check-In Assistant	Logistics

COMPLETION_STATUS	TASK_NAME	EVENT_NAME	ROLE_NAME	SUPPORT_TYPE
In Progress	Distribute Flyers	100m Sprint Final	Flyer Distributor	Security
In Progress	Guide Attendees	Equestrian Jumping	Attendee Guide	Volunteer Coordination
In Progress	Manage Registration	Men Basketball Final	Registration Desk Manager	Catering

COMPLETION_STATUS	TASK_NAME	EVENT_NAME	ROLE_NAME	SUPPORT_TYPE
Completed	Coordinate Volunteers	Gymnastics All-Around	Volunteer Coordinator	Technical
Completed	Set Up Venue	Opening Ceremony	Venue Setup	Medical
Completed	Technical Support	Rowing 8s Final	Technical Support	Media

Figure 3.3: Tables depicting tasks filtered by completion status

Our findings from PowerBI help not only with overall resource allocation by capacity but also track the progress of individual tasks and identify areas for improvement. Figure 3.3 depicts the completion status of each task, along with the event it is associated with and the corresponding role and support type, allowing for the volunteer management system to focus attention towards tasks or roles with common “pending” or “in progress” status. Real-time monitoring allows for swift and continual addressing of bottlenecks in the process, and anchors the management system process within the realities of tasks by drilling them down and linking them to specific events. This is a cornerstone to increase the efficiency of processes, as tasks are broken down and any blockers are addressed.

Hence, the capabilities of PowerBI enable a clearer, data-driven understanding of task progress and volunteer engagement. Visualisations and analysis stemming from them stress the viability of this system, as it tracks statuses and proactively optimises operations, maximising the efficiency of the volunteer management system.

Conclusion

Hence, the Volunteer Management System requires a robust and scalable database system to promote efficient management of volunteers throughout the Olympic Games. The modelling and creation of a data warehouse allows us to gain integral insights into facilitating the ongoing success of this system.

References

BBC News. *Paris 2024: How is France preparing for the Olympics and Paralympics?* (2024) *BBC News*. Available at: https://www.bbc.com/news/world-europe-68018536 (Accessed: 28 July 2024).

Contreras, C. (2024) *When and where are the Paris 2024 Olympics? , NBC Insider Official Site*. Available at: https://www.nbc.com/nbc-insider/where-are-the-olympics-2024-paris-games-sites (Accessed: 28 July 2024).

Dixon, E. (2023) *Inside the event management technology delivering for the Commonwealth Games and the Olympic Movement*, *SportsPro*. Available at: https://www.sportspromedia.com/insights/analysis/wiz-team-sports-event-management-technology-commonwealth-games-olympic-movement/ (Accessed: 28 July 2024).

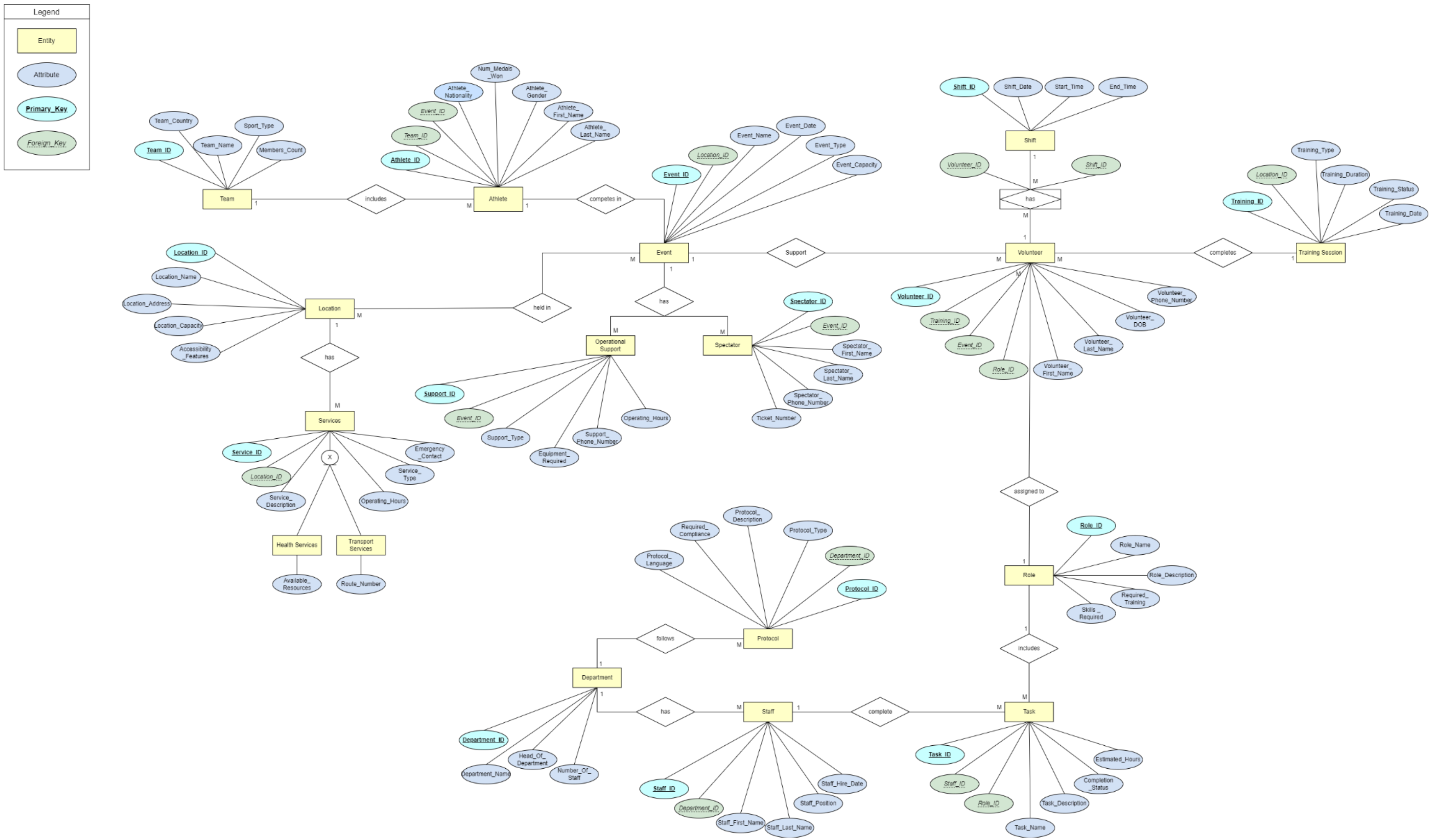
Leuzzi, J. (2024) *What sports are at the 2024 Paris Olympics? , Detroit Free Press*. Available at: https://www.freep.com/story/sports/olympics/2024/07/26/2024-paris-olympics-events-sports-what-how-many-breaking-summer-debut/74271101007/ (Accessed: 27 July 2024).

Paris 2024 volunteer programme (2024) *Olympics*. Available at: https://olympics.com/en/paris-2024/the-games/volunteers (Accessed: 28 July 2024).

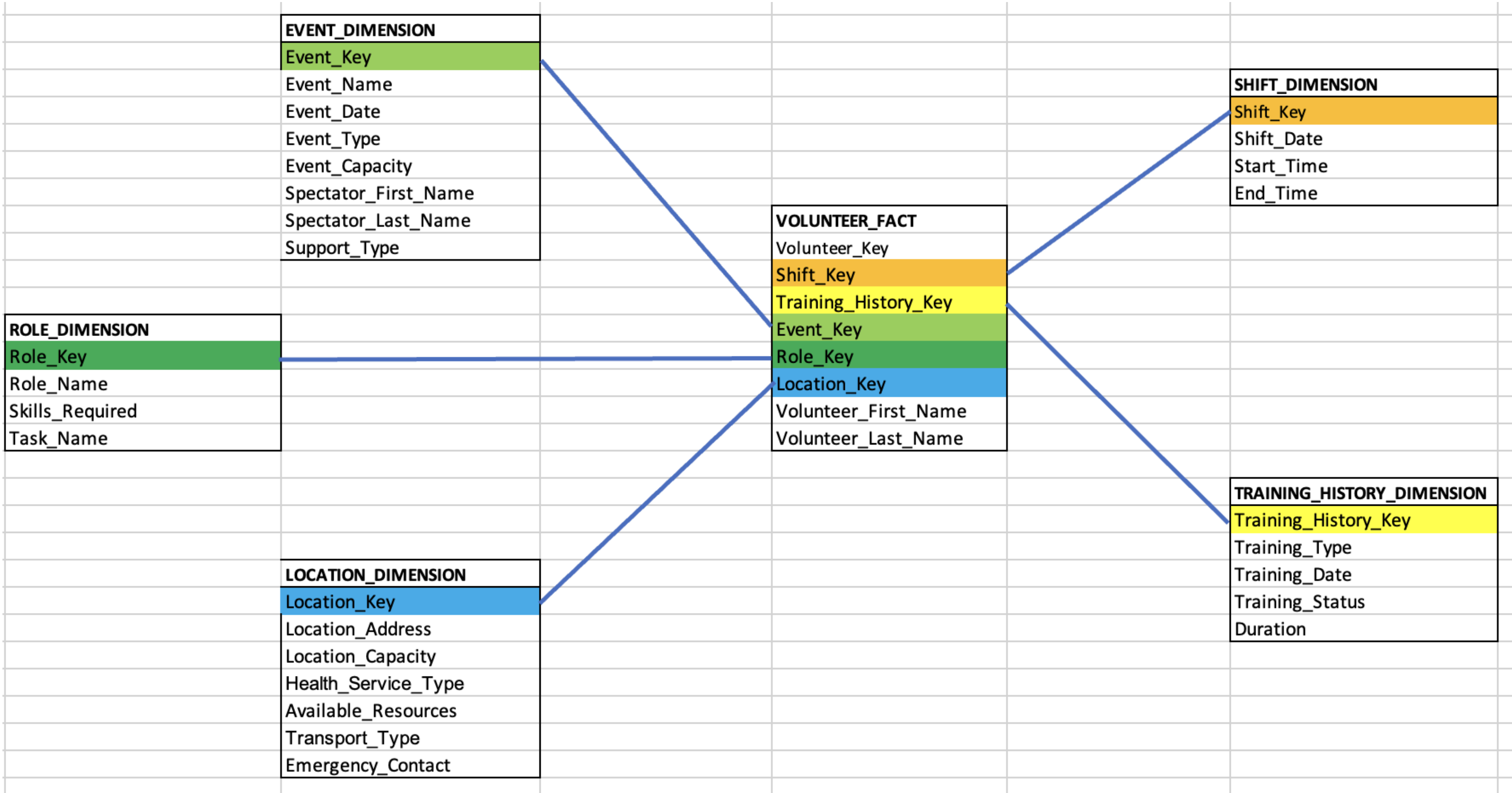
Pop, C. (2013) 'The modern Olympic Games–a globalised cultural and sporting event', *Procedia - Social and Behavioral Sciences*, 92, pp. 728–734. doi:10.1016/j.sbspro.2013.08.746.

Suliman, A. and Granados, S. (2024) *How many athletes, countries are in the 2024 Olympics, and other key numbers*, *How many athletes are in the Paris Olympics, and other key numbers*. Available at: https://www.washingtonpost.com/sports/olympics/2024/07/27/paris-olympic-games-2024-how-many/ (Accessed: 28 July 2024).

Entity-Relationship Model



Star Schema


EVENT_DIMENSION
Event_Key
Event_Name
Event_Date
Event_Type
Event_Capacity
Spectator_First_Name
Spectator_Last_Name
Support_Type
SHIFT_DIMENSION
Shift_Key
Shift_Date
Start_Time
End_Time
VOLUNTEER_FACT
Volunteer_Key
Shift_Key
Training_History_Key
Event_Key
Role_Key
Location_Key
Volunteer_First_Name
Volunteer_Last_Name
ROLE_DIMENSION
Role_Key
Role_Name
Skills_Required
Task_Name
TRAINING_HISTORY_DIMENSION
Training_History_Key
Training_Type
Training_Date
Training_Status
Duration
LOCATION_DIMENSION
Location_Key
Location_Address
Location_Capacity
Health_Service_Type
Available_Resources
Transport_Type
Emergency_Contact